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# Emerging chalcogenide based van der Waals heterostructures for ultrathin excitonic solar cells with enhanced photo-conversion efficiency†

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The emerging chalcogenide based two-dimensional materials, zirconium-diselenide ($ZrSe_2$) and hafnium-diselenide ($HfSe_2$), have been reported to construct promising van der Waals (vdW) heterostructures, which could improve the efficiency of ultrathin excitonic solar cells. We perform systematic first-principles electronic structure calculations to explore the electronic band structures and optical properties and hence determine the power conversion efficiency (PCE) of the heterostructures. We have considered both the 2H and 1T phases of $ZrSe_2$ and $HfSe_2$ which give rise to four different possible heterostructures. From the analysis of band edge alignment, three of the heterostructure configurations have been identified to exhibit type II band alignment, which helps in boosting the effective separation of electron–hole pairs, paving the way towards their application in ultrathin excitonic heterojunction solar cells. The electrostatic potential shows a huge difference in the dip, which signifies the in-built electric field that is generated between two different surfaces, thus enhancing the transfer of electrons and holes between different layers. A greater range of incoming solar energy can be absorbed by the heterostructure, therefore providing better utilization of the incident radiation. The results show that the heterostructure formed by the 2H phase of $ZrSe_2/HfSe_2$ has reached a maximum power conversion efficiency of more than 21%.

## 1 Introduction

The successful experimental realization of graphene[1] paved the way towards technology that utilises two-dimensional (2D) layered materials. Similar to graphene, various other 2D materials such as antimonene,[2] phosphorene,[3] germanene,[4] silicene,[5] hexagonal boron nitride,[6] metal monochalcogenides (MMCs),[7–9] transition metal dichalcogenides (TMDs),[10–13] transition metal trichalcogenides (TMTs),[14–17] MXenes (transition metal nitrides and carbides),[18–20] and Janus materials[21–23] have been investigated, both experimentally and theoretically from the perspective of energy and optoelectronic properties. These 2D materials show a wide range of electronic properties, starting from large band gap insulators[24] to semiconductors,[25] and semimetals.[26] The 2D monolayers have high specific surface area[27] and atomic-scale thickness,[28] and have gained significant interest due to their rare electronic,[29] optical,[30] and thermal properties.[31] Strong in-plane covalent bonds allow these 2D monolayers to tolerate mechanical deformation to a significant extent and provide exceptional mechanical stability, leading to a large elastic limit.[32] On the other hand, the weak van der Waals (vdW) interaction between the layers enables the exfoliation of monolayers from the bulk and the ability to stack the monolayers in the form of vertical heterostructures, since they lack the dangling connection.[33–43] This gives us the opportunity to expand the library of 2D materials and consider them for emerging applications based on heterostructures formed of 2D materials. These vdW heterostructures can be used in various solid-state devices, in diverse applications including solar cells,[33–35] photodetectors,[36,37] photocatalysts,[38,39] semiconductor lasers,[40,41] and light-emitting diodes (LEDs).[42]

The TMDs are the emerging class of 2D materials having the general formula $MX_2$, where M is the transition metal and X is the chalcogens such as S, Se, and Te. The TMDs form a lamellar structure with the sequence X–M–X, where two layers of chalcogen atoms surround the metal atoms. Most commonly, TMDs exist in two phases *i.e.*, 2H and 1T phases.[44] In TMDs the electronic, optical, and transport properties are highly influenced by the phase in which they exist, even after having the same chemical composition. For example, the 2H phase of the $MoS_2$ monolayer, which is more stable in ambient conditions, has a semiconducting nature with a band gap of 1.2 eV.[45,46] The $MoS_2$ monolayer's 1T phase, which has been synthesized and investigated, is a strong contender for catalytic activity, has a no band gap and exhibits metallic behaviour.[45,47] The hexagonal 2H phase contains the chalcogen in trigonal prismatic

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coordination ($D_{3h}$), which surrounds the transition metal atom. Group VI of the TMDs which includes $MoS_2$, $WSe_2$ and so on are most commonly found to be in this phase in ambient conditions. With octahedral coordination ($D_{3d}$), the 1T phase has a single layer stack with tetragonal symmetry. Under normal circumstances, group IV of the TMDs is most frequently observed to be in this phase. The structural transitions between the 1T and 2H phases can take place through geometric rotations.[48] The transversal displacement of one chalcogen layer induces the phase transition from 2H to the 1T phase. In addition to their great chemical stability, these materials exhibit excellent mechanical characteristics and a low-cost easier synthesis, making them promising for various applications. Owing to these advantageous properties of TMDs, rigorous effort has been invested both in terms of experimental and theoretical investigation, to find better TMDs-based materials through several design techniques. The synergistic effect of individual monolayers has been manifested in the form of the heterostructure, to improve functionality, stability and device efficiency.

The sustainability problem of rising global energy demand could be effectively addressed through solar photovoltaics (PV) technology, which directly converts solar energy into electricity. PV technology is environmentally friendly, inexpensive, and easy to implement for practical application. In the last decades, traditional silicon (Si)-based solar cells have been utilized by industry due to their low manufacturing cost.[49] However, in the last few years, following their inception, hybrid perovskites have been explored by many groups for their application in photovoltaic devices.[50,51] This class of materials has reached similar efficiency, which is comparable to Si-based solar cells.[52] Nonetheless, there is still a demand for thin and flexible PV materials with improved mechanical stability and the capacity to tolerate severe mechanical stress. In this context, 2D materials are more appropriate because they have a high power conversion rate per weight.[53] Extensive effort has been invested into demonstrating and implementing 2D materials with remarkable optoelectronic applications, specifically beneficial for light-to-energy conversion in photovoltaic systems. Under illumination, the split metallic d–d band electronic structure of the TMDs class of 2D materials remains stable, ensuring a longer life.[54] The TMDs also have high carrier mobility, which is also crucial in designing photovoltaic devices.[55–57] As the materials used for photovoltaic technology should absorb a broad spectrum of incident solar light, the materials that are best suitable for such an application should preferably have a band gap in the range 1–2 eV. In this regard, the TMDs have a wide range of bandgaps, among which $ZrSe_2$ and $HfSe_2$ possess a nominal bandgap that is almost similar to Si for this application, and they also have a small lattice mismatch, making them ideal for ultrathin excitonic solar cells. The corresponding thin films of these materials in the 1T phase have been synthesised and investigated by experiments in earlier works.[58,59] Furthermore, the synthesis of the 2H phase is favored by the phase transition between the 1T and 2H phases that has already been seen, which has been researched less.[60,61] However, chemical vapour transport has also been utilized to synthesize the 2H phase of $HfSe_2$, which has been utilized in other applications like field effect transistors;[62] this also paves the way for the synthesis of $ZrSe_2$ in the 2H phase. The efficiency of photovoltaic devices can be further enhanced as electrons and holes get separated and transported to their respective transport layers, reducing recombination losses, which can be done using the formation of heterostructures. Recently, computational work has been carried out to demonstrate 2D heterostructures as excitonic solar cells[35,63–67] and also Schottky barrier solar cells,[68] for these classes of materials.

In this work, we perform a systematic series of computational investigations of 2D monolayer $MSe_2$ (M = Zr, Hf) in both 1T and 2H phases. The individual monolayers are stacked in an energetically more advantageous way to create vdW heterostructures. The stability factors, such as thermodynamic and mechanical stability, along with stability at higher temperatures, were noted for both the monolayers and heterostructures. Throughout this work, we use density functional theory (DFT) to calculate the electronic structure and optical properties of the heterostructures formed by the 1T and 2H phases of $ZrSe_2$ and $HfSe_2$, and determine their power conversion efficiency (PCE) to consider them as possible candidates for excitonic solar cells.

## 2 Computational methodology

We have employed Vienna *ab initio* simulation software (VASP)[69,70] within the framework of density functional theory (DFT) formalism[71,72] for all first principles calculations. The electron–ion interactions are best described by the projector-augmented wave (PAW)[73,74] formalism, while throughout this study, the Perdew–Burke–Ernzerhof (PBE) form of the generalized gradient approximation (GGA)[75] has been used as the exchange–correlation functional. Due to the presence of a heavy element in the monolayers, the spin-orbit coupling (SOC)[76] effect has been taken into account. An adequate vacuum of at least 15 Å is added along the $z$ direction to avoid interaction between the adjacent layers. The correction to van der Waals (vdW) interactions is introduced using the DFT-D3 method.[77] The converged energy cut-off of 500 eV is considered for the plane wave basis sets throughout the calculations. We have considered complete lattice relaxation in all the structures until the Hellman–Feynman forces of continuous atoms become less than $2 \times 10^{-3}$ eV Å$^{-1}$ and the energy convergence was achieved with a convergence factor of $1 \times 10^{-5}$ eV. The 2D Brillouin zone has been sampled with a Monkhorst–Pack[78] $k$-point mesh of $11 \times 11 \times 1$ for the unit cell throughout the calculation after performing the convergence test. Complete lattice relaxation is performed while constructing the heterostructure by stacking the monolayers with varying distances and stacking orders (shown in ESI Fig. 1†) to obtain the structure with minimum energy configuration, which is considered in further calculations. In order to obtain a band gap with higher precision, we have considered the Heyd–Scuseria–Ernzerhof hybrid functional (HSE06) type of exchange correlation functional for the calculation of electronic and optical properties.[79] Bader charge transfer calculations are performed[80] to understand the bonding scenario between the constituent elements. The elastic

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constants for understanding the mechanical stability are derived from the variation of energy with respect to strain, corresponding to the uniaxial and biaxial deformations and considering the second derivative of these energies. The relation has been obtained by varying the magnitude of the strain in steps of 0.5%, ranging from −2% to 2%. Dynamic stability calculations were performed using the finite-displacement method, as implemented in the Phonopy package.[81] The thermal stability of the structures are considered using *ab initio* molecular dynamics simulations at 300 K by constructing a 2 × 2 × 1 supercell. The canonical ensemble (NVT) with the Nose–Hoover thermostat,[82,83] is considered for a total simulation time 10 ps with a time step of 1 fs.

# 3 Results and discussion

## 3.1 Crystal structure and stability

**3.1.1 Monolayers.** The unit cell of $MSe_2$ (M = Hf, Zr) consists of three atomic layers in the vertical sequence Se–M–Se. These monolayers are arranged in two well-known phases similar to the other TMD monolayers.[84,85] The octahedral (1T) phase belongs to the $P\bar{3}m2$ space group and hexagonal (2H) phase belongs to $P\bar{6}m2$ space group. The bird's eye view and the side view of the 1T and 2H phases are shown in Fig. 1(a)–(d), and their first Brillouin zone is demonstrated in Fig. 1(e).

We have performed complete lattice relaxation individually for all the considered systems, to obtain the optimized lattice parameters $a$, as shown in Table 1 which are consistent with previously reported results.[86–88] Compared to the 1T phase, the 2H phase's lattice parameter is marginally smaller. $HfSe_2$ has a slightly smaller lattice parameter than $ZrSe_2$ in the same phase; this is due to the slightly shorter bond length of Hf–Se than Zr–Se. The bond angle at the selenide atom $\theta_1$ varies proportionally to the lattice constant $a$ and is marginally higher in the 1T phase; the angle $\theta_2$ at the transition metal atom is higher in the 1T phase which varies with $a$. The thermodynamic stability of the monolayer is determined using the formation energy eqn (1) as follows:

$$E_{\text{formation}} = E_{\text{MSe}_2} - \mu_{\text{M}} - 2\mu_{\text{Se}} \tag{1}$$

where $E_{\text{MSe}_2}$ is the total energy of the monolayer, $\mu_{\text{M}}$ and $\mu_{\text{Se}}$ are the chemical potentials of the transition metal atom and the selenium atom, respectively. All the constituent monolayers are thermodynamically stable while exhibiting negative formation energy. The 1T phase in both the monolayers is found to be more stable, considering that they have a lower formation energy compared to the 2H phase. Further, the mechanical stability is investigated from the variation of energy with respect to strain, by taking the second derivative to obtain the elastic constants, as follows:

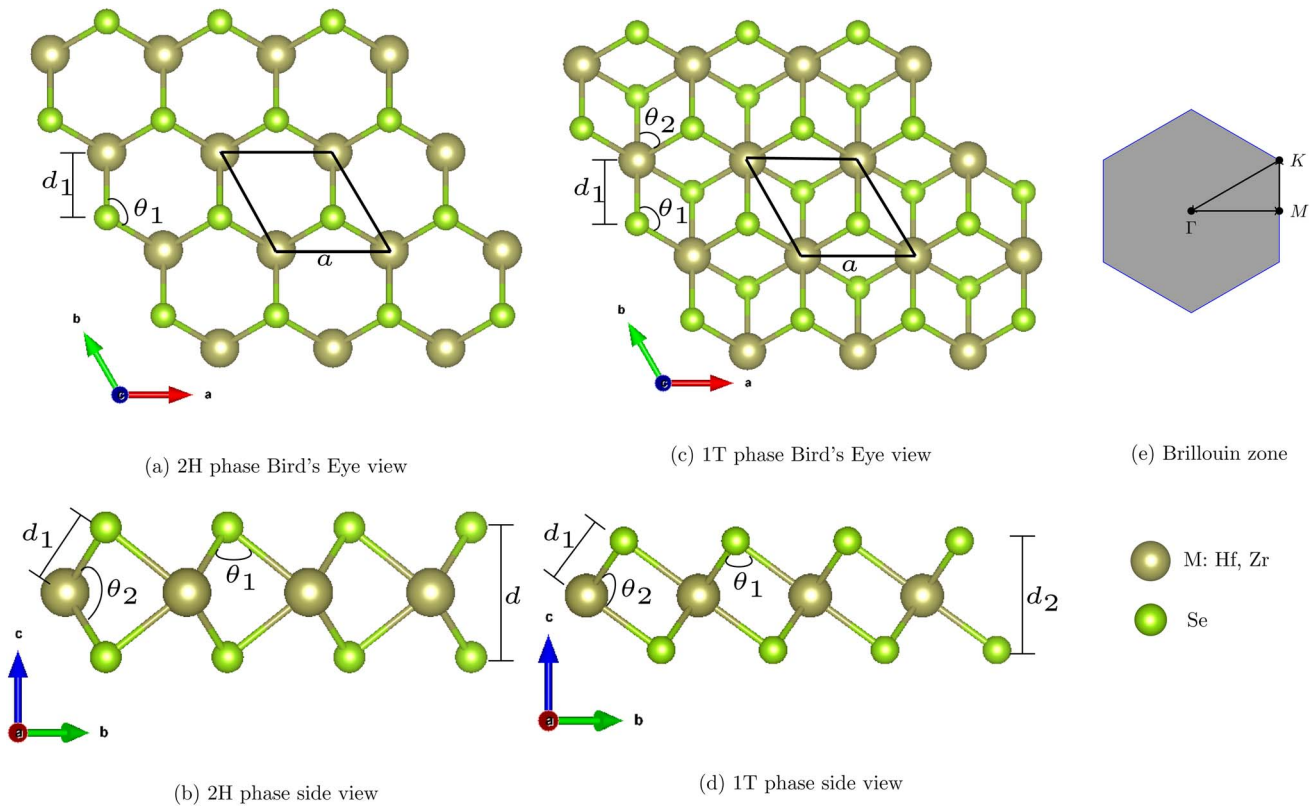


Fig. 1 Geometric structure of 2H and 1T phase TMD monolayers ($MSe_2$; M = Hf, Zr). (a and b) Bird's-eye view and side view of the 2H phase. (c and d) Bird's-eye view and side view of the 1T phase. (e) Brillouin zone for the hexagonal cell. $d_1$ is the distance between (Hf/Zr) and Se, $d_2$ is the thickness of the monolayer, $\theta_1$ is the angle formed at the selenide atom and $\theta_2$ is the angle formed at the metal atom.

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Table 1 Lattice constant $a$ of the unit cell; bond length between the transition metal atom and Se, $d_1$; layer heights $d_2$; bond angles $\theta_1$, $\theta_2$; formation energy, $E_{\text{formation}}$; elastic constants $C_{11}$ and $C_{12}$; charge transferred by M ($q_M$) and charge received by Se ($q_{Se}$)

| Monolayer | | $a$ (Å) | $d_1$ (Å) | $d_2$ (Å) | $\theta_1$ | $\theta_2$ | $E_{\text{formation}}$ (eV) | $C_{11}$ (N m$^{-1}$) | $C_{12}$ (N m$^{-1}$) | $q_M$ (e$^-$) | $q_{Se}$ (e$^-$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $HfSe_2$ | 2H | 3.66 | 2.70 | 3.36 | 85.24° | 77.12° | −3.15 | 81.89 | 27.17 | 1.87 | −0.94 |
| | 1T | 3.75 | 2.68 | 3.16 | 88.80° | 91.20° | −3.66 | 72.54 | 13.92 | 1.78 | −0.89 |
| $ZrSe_2$ | 2H | 3.69 | 2.72 | 3.38 | 85.42° | 76.89° | −3.25 | 73.90 | 26.44 | 1.59 | −0.80 |
| | 1T | 3.79 | 2.70 | 3.18 | 88.97° | 91.02° | −3.67 | 67.13 | 13.09 | 1.76 | −0.88 |

$$C_{11} = \frac{1}{A_0}\frac{\partial^2 E_u}{\partial s^2}\bigg|_{s=0}, \quad 2(C_{11}+C_{12}) = \frac{1}{A_0}\frac{\partial^2 E_b}{\partial s^2}\bigg|_{s=0} \tag{2}$$

where, $E_u$ and $E_b$ are the total energies when subjected to uniaxial and biaxial strain, respectively, and $A_0$ is the surface area. All the monolayers satisfy the condition for Born–Huang mechanical stability for hexagonal monolayers ($C_{11} > 0$ and $C_{11} > |C_{12}|$) suggesting that the monolayers are mechanically stable.[89,90] Calculating the phonon band structures of each monolayer revealed that they were all dynamically stable, as seen in Fig. 2 The dynamic stability of the monolayers is confirmed by the absence of negative frequencies in the phonon band dispersions. The total energy fluctuation of the monolayers for 10 ps at 300 K is shown in Fig. 3 and the thermal stability is consistently confirmed by the time-dependent evolution of the total energy. It is clear from the snapshots of the structure taken both before and after annealing (Fig. 3, inset), that there is no substantial structural reconstruction at 300 K.

**3.1.2 Heterostructures.** The heterostructure is formed by stacking the optimized monolayers one above the other in the form of bilayers. The lattice mismatch between the monolayers, denoted as $\Delta$, which is determined using the relationship described in eqn (3), is found to be less than 3.5%. Therefore the construction of a heterostructure based on their unit cells, would ensure that there is only a small amount of strain exerted on the monolayers. Different stacking arrangements, such as AA, AB, AB′, and AB″, (shown in ESI Fig. 1†) are considered for different combinations of monolayers by varying the distances between them to obtain heterostructures with the lowest possible minimum energy. Four heterostructures with different combinations of monolayers are considered for further calculation. We have obtained the binding energy $E_b$ as given by eqn (4). These binding energies have magnitudes similar to those of the other heterostructures reported in previous studies, signifying that the heterostructure can be formed.[91] The lattice constants, stacking order, and interlayer distance after optimization are depicted in Table 2. In comparison to AB and AB′, the heterostructure with AB″ stacking has a smaller interlayer spacing, optimizing the spatial separation between the monolayer's closest atoms. The heterostructure's optimized lattice constant is between each individual monolayer's lattice constant, but leans toward the monolayer with the lower lattice constant. As a result, higher compressive strain is applied to the monolayer with the greater lattice constant. The elastic constant is determined as shown in eqn (2), which meets the conditions for mechanical stability. The energy fluctuation of the four different combinations of heterostructure over 10 ps at 300 K is shown in Fig. 4. The confined range of fluctuations in the time-dependent evolution of total energy reliably verifies the corresponding thermal stability. The snapshots of the structure (Fig. 4, inset) are taken before and after 10 ps make it quite evident that no structural reconstruction takes place at 300 K temperature.

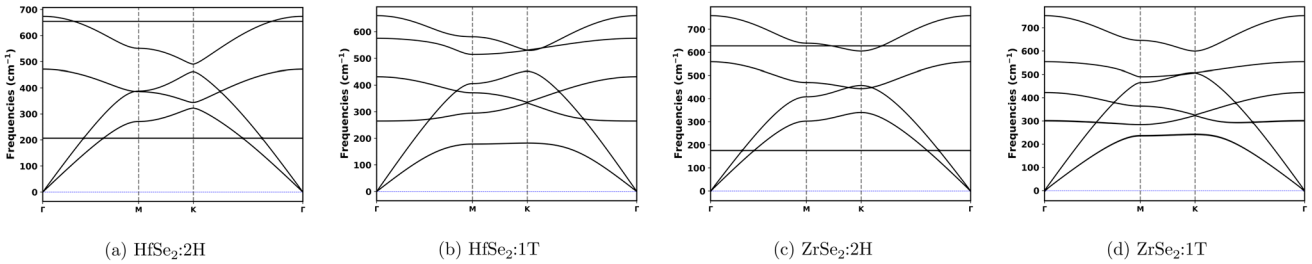
$$\Delta = \frac{\left|a_{ZrSe_2} - a_{HfSe_2}\right|}{a_{ZrSe_2}} \times 100 \tag{3}$$

where $\Delta$ is the lattice mismatch, $a_{ZrSe_2}$ and $a_{HfSe_2}$ are the lattice parameters of $ZrSe_2$ and $HfSe_2$ monolayers, respectively, forming the hetero-bilayers.

$$E_b = E_{ZrSe_2+HfSe_2} - (E_{ZrSe_2} + E_{HfSe_2}) \tag{4}$$

where $E_b$ is the binding energy, $E_{ZrSe_2+HfSe_2}$ is the total energy of the heterostructure and $E_{ZrSe_2}$, $E_{HfSe_2}$ are the total energy of each individual monolayer.

To have a better understanding of the interaction between layers in the heterostructure, we determined the difference in charge density ($\Delta\rho$), shown in Fig. 5, which is defined as the difference in charge density of the heterostructure $\rho_{(ZrSe_2/HfSe_2)}$




(a) $HfSe_2$:2H (b) $HfSe_2$:1T (c) $ZrSe_2$:2H (d) $ZrSe_2$:1T

Fig. 2 The phonon band dispersions for the monolayers along the high symmetry points of the first Brillouin zone.

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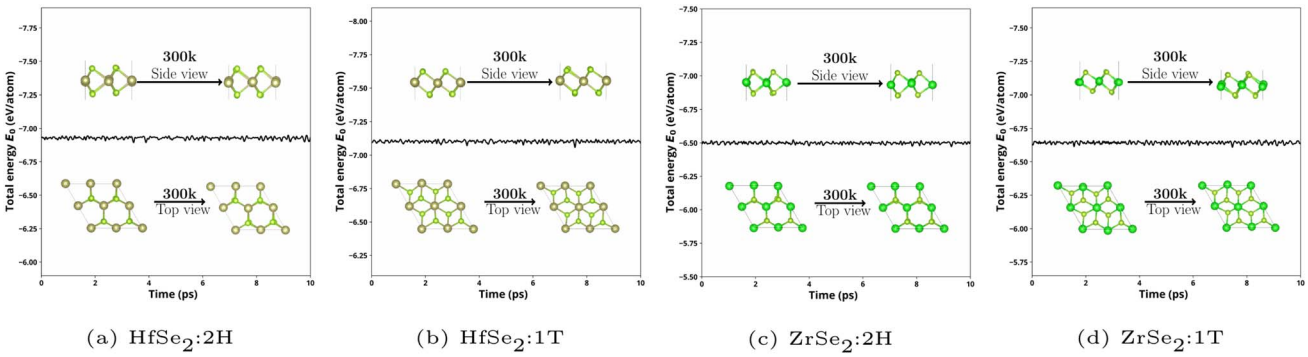


Fig. 3 Time-dependent evolution of the total energy of the monolayers and snapshots at 0 ps and 10 ps, at 300 K.

relative to that of the individual monolayers, as given in eqn (5),

$$\Delta\rho = \rho_{(ZrSe_2/HfSe_2)} - \rho_{ZrSe_2} - \rho_{HfSe_2}. \quad (5)$$

Fig. 5 shows both charge depletion in cyan and charge accumulation in yellow, and the average charge density difference projected along the $c$ direction. The charge density difference at the interface looks similar in all of the above heterostructures, where we have charge accumulation at the center, which is surrounded by a charge depletion region near the interface. However, the charge density difference at the interface varies depending on the phase of the monolayers; for the 2H phase, there is small charge accumulation at the interface, and for the 1T phase, it is closer to zero for AB″ stacking and slightly negative for AB′ stacking.

### 3.2 Electronic properties

**3.2.1 Monolayers.** We have systematically investigated the electronic properties of the monolayers through the determination of projected density of states (PDOS) and the electronic band structures, Fig. 6. The SOC effect has been taken into account during the electronic structure calculations. The GGA-PBE calculations underestimate the band gap to a large extent, as shown in the SI (Fig. 7†). Therefore, we have used the HSE06 type hybrid exchange correlation functional, which estimates the band gaps closer to the experimental values. The estimated band gaps for the monolayers are given in Table 3 and are found to be between 0.97 eV and 1.53 eV, which lies near the infrared region showing that they absorb significant amounts of solar radiation, crucial for efficient solar cell applications. The 2H phase of monolayers is found to have a larger band gap as compared to the 1T phase, and the difference between the band gaps of $HfSe_2$ and $ZrSe_2$ of the same phase is relatively small. A deeper understanding of the partial DOS (PDOS) and the band structure reveals that the Se-4p orbital dominates the conduction band (CB) with a small contribution from the Zr-4d and Hf-5d orbitals in the 2H phase. In the CB of the 1T phase, the contribution is solely due to the Se 4p orbital. In the valence band (VB) regime, the main contribution in all monolayers, is from the outermost Zr-4d and Hf-5d orbitals in $ZrSe_2$ and $HfSe_2$, respectively. From the band structure, we can observe that the monolayers have an indirect band gap. In the 1T phase of both monolayers, the conduction band minima (CBM) lies at the $M$ (0.5,0,0) and the valence band maxima lies at the $\Gamma$ (0,0,0). Whereas in the 2H phase, there is a slight shift in the position of the CBM from the high symmetry point ($M$) towards $K$ (0.33,0.33,0) in $HfSe_2$ and towards $\Gamma$ in the case of $ZrSe_2$. The valence band maxima (VBM) for the 2H phase lies between $\Gamma$ and $M$ for $ZrSe_2$ and $K$ and $\Gamma$ for $HfSe_2$.

The carrier mobility plays an important role in the performance of photovoltaics. Therefore, we have estimated the longitudinal acoustic phonon-limited mobility of electrons and holes in the monolayers, based on deformation potential (DP) theory. This is a widely employed theory used to calculate the carrier mobility in 2D materials suggested by Bardeen and Shockley.[92] According to this theory, the charge carrier mobility mainly depends on the DP constant of the charge carriers, effective mass, and elastic constant $C_{11}$, as expressed using eqn (2). Generally, the mobility, which varies inversely with the square of the DP constant, is defined as the slope of the variation in band edge position (VBM for holes and CBM for electrons) with infinitesimal strain and can be expressed as

Table 2 Lattice constant of the hetero-bilayer $a$; lattice mismatch $\Delta$ between the layers present in the heterostructure; $t$ interlayer distance; $E_b$ binding energy; elastic constants $C_{11}$ and $C_{12}$

| Heterostructure | $a$ (Å) | $\Delta$ (%) | $t$ (Å) | Stacking | $E_b$ (meV) | $C_{11}$ (N m$^{-1}$) | $C_{12}$ (N m$^{-1}$) |
|---|---|---|---|---|---|---|---|
| $ZrSe_2$:2H/$HfSe_2$:2H | 3.67 | 0.81 | 3.16 | AB | −0.25 | 153.29 | 53.87 |
| $ZrSe_2$:1T/$HfSe_2$:2H | 3.71 | 1.62 | 3.05 | AB″ | −0.23 | 145.59 | 40.53 |
| $ZrSe_2$:2H/$HfSe_2$:1T | 3.71 | 3.43 | 3.14 | AB′ | −0.24 | 143.29 | 42.72 |
| $ZrSe_2$:1T/$HfSe_2$:1T | 3.76 | 1.05 | 3.07 | AB″ | −0.27 | 138.79 | 25.30 |

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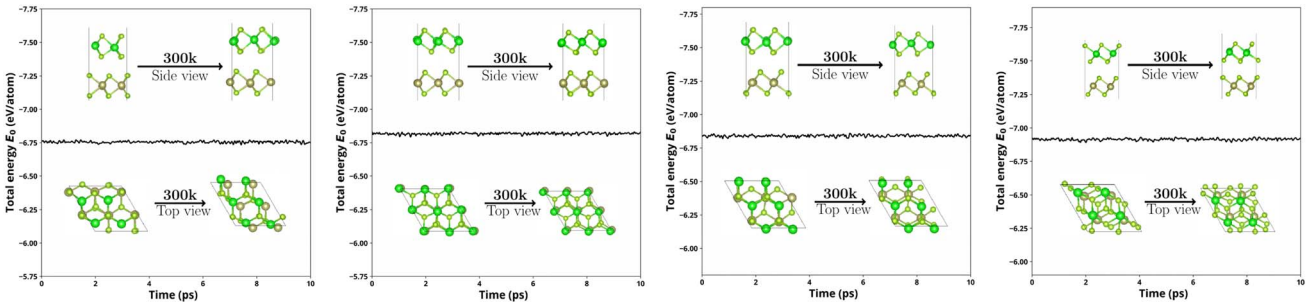


(a) $ZrSe_2$:2H/$HfSe_2$:2H (b) $ZrSe_2$:1T/$HfSe_2$:2H (c) $ZrSe_2$:2H/$HfSe_2$:1T (d) $ZrSe_2$:1T/$HfSe_2$:1T

Fig. 4 Evolution of total energy of the heterostructure and snapshots at 0 ps and 10 ps at 300 K.

$E_x = \frac{\partial E_{edge}}{\partial \varepsilon}$; where $\varepsilon$ is the applied uniaxial strain defined by $\varepsilon = (a_{str} - a_0)/a_0$ ($a_{str}$ and $a_0$ are the lattice constants with strain and without strain, respectively). $E_{edge}$ is the shift in the conduction or valence band edge position for electrons and holes respectively. Considering this, the mobility is expressed as:

$$\mu_{2D} = \frac{e\hbar^3 C_{11}}{k_B T m_1^* m_d E_x{}^2}. \quad (6)$$

Here, $m^*$ is obtained using $\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E(k)}{dk^2}$, $m_1^*$ is the effective mass of the electron or holes in the transport direction and $m_d = \sqrt{m_1^* m_2^*}$ is the average effective mass, $k_B$ is the Boltzmann constant, and $T$ is room temperature (300 K). It can be noted that both monolayers in the 1T phase have higher mobility of electrons and holes, whereas the 2H phase has lower mobility; this is due to the flattened band, giving rise to high effective mass and also a comparably larger change in band edge position with the presence of strain, which results in higher deformation potential.

**3.2.2 Heterostructures.** To obtain a more profound understanding of the electronic band structure in the heterostructure, the hybrid-HSE06 exchange–correlation functional has been employed. Fig. 7 shows the electronic properties that have been investigated using the atomic layer projected band structure. In Fig. 7(a)–(c) we observe that the valence band maximum of one material is positioned between the valence band and conduction band of another material, indicating a type-II band alignment. In the case of the heterostructure formed by stacking $HfSe_2$:1T and $ZrSe_2$:1T (Fig. 7(d)), the band edges of $ZrSe_2$ lie between the band edges of $HfSe_2$, which signifies the formation of type-I band alignment. The PDOS shows that the Se atom plays a significant role in the valence band region, which is similar to the case of the monolayers. Se atoms from $ZrSe_2$ contribute near the VB region in the heterostructure shown in Fig. 7(a), (b) and (d), and the Se atoms of $HfSe_2$ contribute in the case of $ZrSe_2$:2H/$HfSe_2$:1T. In the conduction band region, the metal atom contributes mainly near the Fermi level, which is also similar to the monolayer case.

The mobility of the charge carriers in the heterostructures are calculated using DP theory and shown in Table 1 of the ESI.† In contrast to having a higher elastic constant, $C_{11}$, which is directly proportional to mobility, the DP constant was also found to be high, which is inversely proportional therefore affecting only small changes in mobility. Nonetheless, it has been discovered that hole mobility drastically increases upon heterostructure construction. The band gaps determined for all the heterostructures are found to be indirect, lying between 0.35 eV and 1.23 eV. It could be noted that the heterostructures formed by stacking with a different phase of the material have a lower band gap as compared to the ones having the same phase. The reason behind this could be understood from the study of the electrostatic potential shown in Fig. 8. The work function for the 2H phase monolayers is higher as compared to the 1T phase monolayers as depicted in Table 3. The work function for the heterostructures with the same phase shown in Table 4 is similar to their individual monolayers (Table 3). When we consider the heterostructure constructed by considering the different phases, we notice that the work function is almost equal to the one having the lower value, which is the 1T phase. Therefore, we can observe a larger offset that has been created when we consider heterostructures with different phases, which reduces the band gap drastically. Since this heterostructure formed by different phases has a work function close to the work function of the 1T phase monolayer present in the heterostructure, we see that there is a smaller shift in the band edge positions of the 1T phase; whereas the band edge positions of the 2H phase undergo a greater change in the heterostructure constructed by monolayers with different phases. The comparison of the projected band structure of the heterostructure with that of the monolayers reveals that the bandgap of the monolayers in the heterostructure generally decreases slightly, by almost 0.3 eV as compared to the individual monolayers. This indicates that each monolayer in the heterostructure is having a subtle influence on the electronic properties of the others. Fig. 8 also shows a large dip in electrostatic potential difference between $HfSe_2$ and $ZrSe_2$, which is larger in the heterostructure formed by both 2H phases – around 7.68 eV; and there is a slight decrease in the difference when considering the other heterostructures with the lowest being 6.85 eV, which is for the heterostructure formed by the combination of both 1T phases. This difference in electrostatic potential is a crucial factor because it generates an internal built-in electric field between the monolayers, and this field facilitates the transfer of photogenerated charges.

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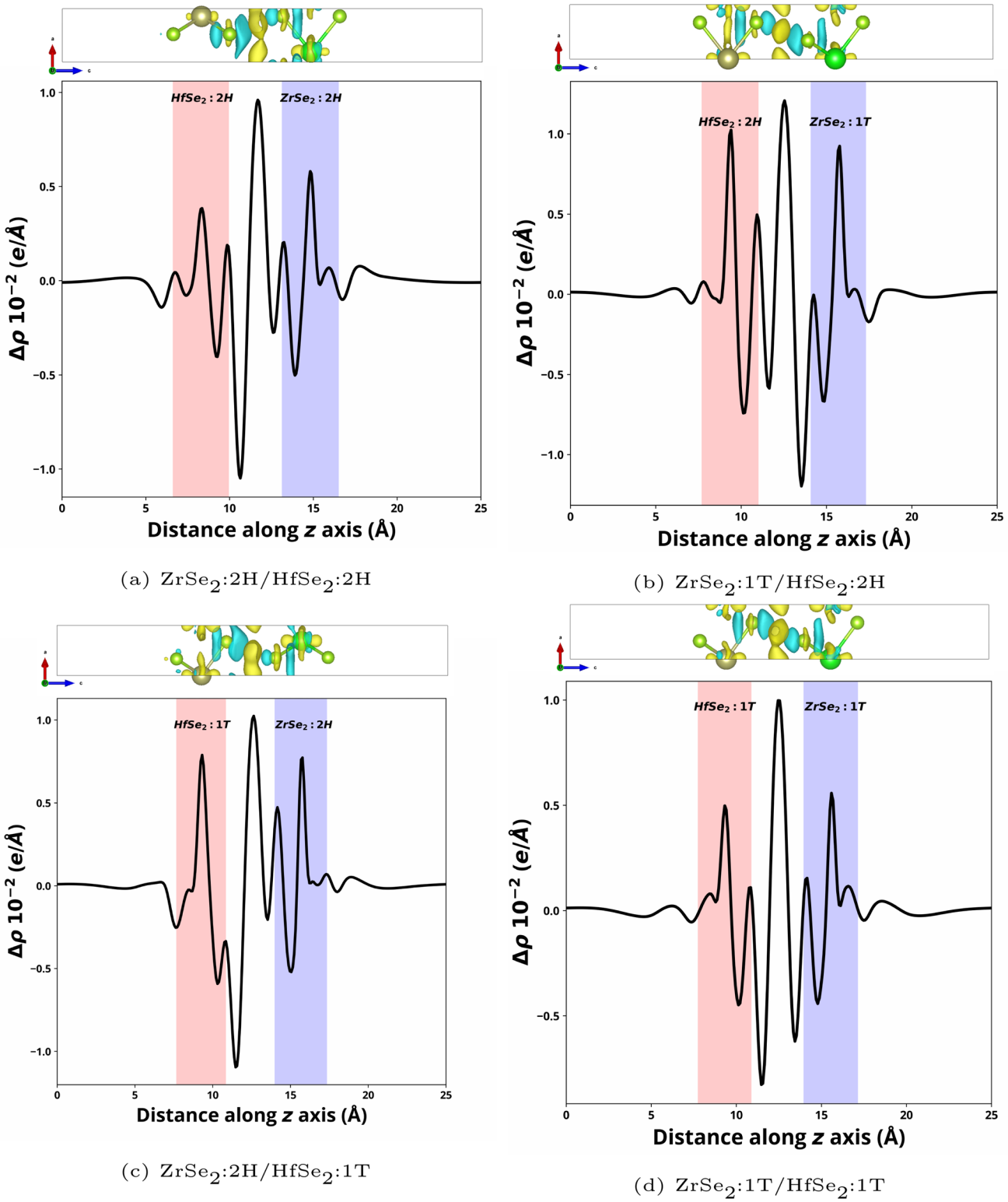


Fig. 5 Charge density difference plot along with the projection in the *c* direction of the different heterostructure configurations. Yellow and cyan correspond to charge accumulation (electron gain) and charge depletion (electron loss), respectively.

### 3.3 Optical properties

The photovoltaic materials must effectively capture the incident solar radiation across the spectrum, particularly in the infrared and visible regions, to maximize the energy conversion efficiency. Therefore, the absorption coefficient, derived from the frequency-dependent complex dielectric function using the

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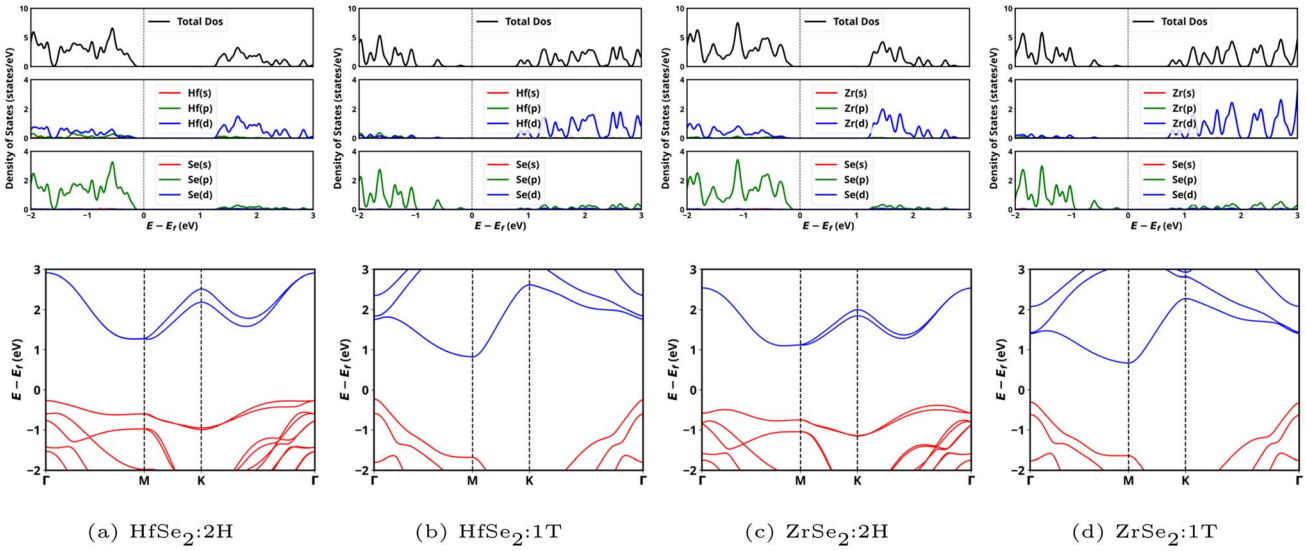


Fig. 6 The top section shows the projected and total density of states, and the bottom section shows the band structure of the monolayers, determined using the HSE06 functional.

HSE06+SOC, is compared for both monolayers and heterostructures as described in eqn (7),[93]

$$\alpha(E) = \frac{2E}{\hbar c}\left(\frac{\sqrt{\varepsilon_1{}^2 + \varepsilon_2{}^2} - \varepsilon_1}{2}\right)^{\frac{1}{2}} \tag{7}$$

where $\alpha(E)$ is the absorption coefficient measured as a function of incident solar flux, $\varepsilon_2$ is the imaginary part of the complex dielectric function, and $\varepsilon_1$ represents the real part of the complex dielectric function, which is obtained by the Kramer–Kronig transformation.[94] Fig. 9(a) and (b) represent the optical absorption coefficient of all the studied monolayers and heterostructures, along with the spectral irradiance as a function of the photon energy. We observe an absorption coefficient of both the monolayers and heterostructure, widely distributed around the spectral irradiance, which is obtained from the AM1.5 spectra.[95] Meanwhile, we could also observe that in the case of the monolayers, the peak of the absorption coefficient is larger in magnitude by almost two times, than in the 2H phase. The absorption coefficient of the heterostructure exhibits a shift in its peak position, aligning more closely with the peak of the spectral irradiance. This shift indicates that the heterostructure may now absorb light more efficiently at wavelengths where the irradiance is highest. As a result, the material can capture a wider range of incident radiation, improving the overall performance of the solar cell. This enhanced absorption capacity means that the heterostructure can effectively utilize more available incident light, potentially leading to greater efficiency.

### 3.4 Band alignment and power conversion efficiency (PCE) of the heterostructures

Understanding band alignment is a crucial part of designing optoelectronic devices, as it optimizes charge carrier dynamics and improves device performance. During the formation of the heterostructure, there can be three different possibilities of band alignment, namely the straddling alignment (type-I), the staggered alignment (type-II), and the broken alignment (type-III), which are shown in ESI Fig. 8.† The staggered alignment plays a crucial role in enhancing electron–hole separation, which is essential for improving the efficiency of photoconversion. In type-II heterojunctions, the CBM of material 1 (M1) is higher in energy than the CBM of material 2 (M2), and the VBM of M1 is higher than the VBM of M2. This creates an energetic offset at the interface between the two materials. During the light absorption by the heterojunction solar cell, photoexcitation takes place *i.e.*, electrons and holes are generated in both materials. The electrons in the material with the higher CBM (M1) will tend to move to the material with the

Table 3 Band gap $E_g$; work function $\Phi$; deformation potential, $E_{ex}$ and $E_{hx}$, of electrons and holes; effective mass $m^*_{e1}$ and $m^*_{e2}$ of electrons and $m^*_{h1}$ and $m^*_{h2}$ of holes; and mobility $\mu_e$ and $\mu_h$ of electrons and holes respectively in monolayers

| Monolayer | | $E_g$ (eV) | $\Phi$ (eV) | $E_{ex}$ (eV) | $E_{hx}$ (eV) | $m^*_{e1}$ ($m_0$) | $m^*_{e2}$ ($m_0$) | $m^*_{h1}$ ($m_0$) | $m^*_{h2}$ ($m_0$) | $\mu_e$ (cm$^2$ V$^{-1}$ s$^{-1}$) | $\mu_h$ (cm$^2$ V$^{-1}$ s$^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $HfSe_2$ | 2H | 1.53 | 6.68 | −3.01 | −4.68 | 1.01 | 1.20 | −3.00 | −4.38 | 173.12 | 7.32 |
| | 1T | 1.05 | 5.67 | −1.04 | −7.16 | 0.37 | 2.79 | −0.19 | −0.19 | 3799.63 | 834.8 |
| $ZrSe_2$ | 2H | 1.49 | 6.68 | −3.62 | −2.96 | 0.91 | 1.34 | −1.57 | −2.62 | 119.52 | 56.42 |
| | 1T | 0.97 | 5.77 | −1.46 | −7.13 | 0.39 | 1.98 | −0.19 | −0.19 | 1957.11 | 779.05 |

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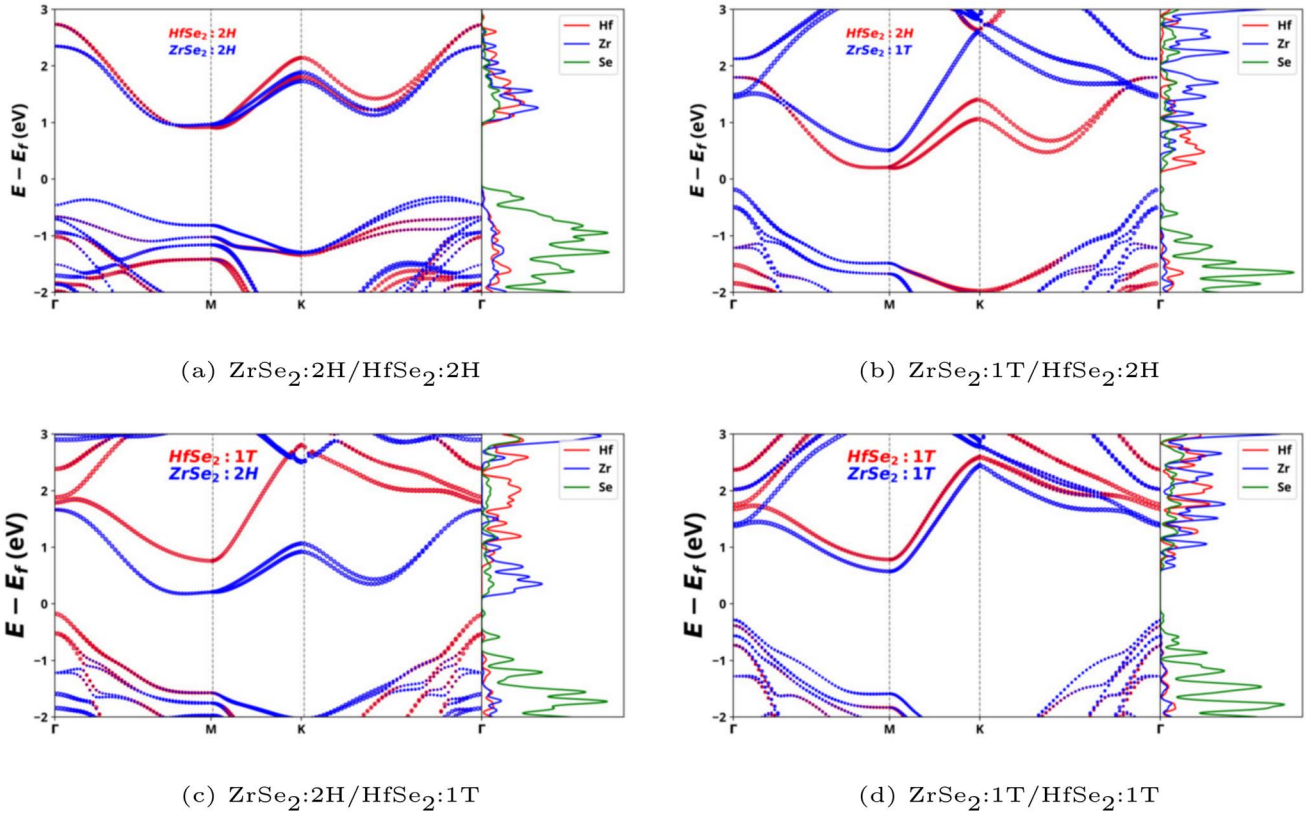



(a) $ZrSe_2$:2H/$HfSe_2$:2H (b) $ZrSe_2$:1T/$HfSe_2$:2H

(c) $ZrSe_2$:2H/$HfSe_2$:1T (d) $ZrSe_2$:1T/$HfSe_2$:1T

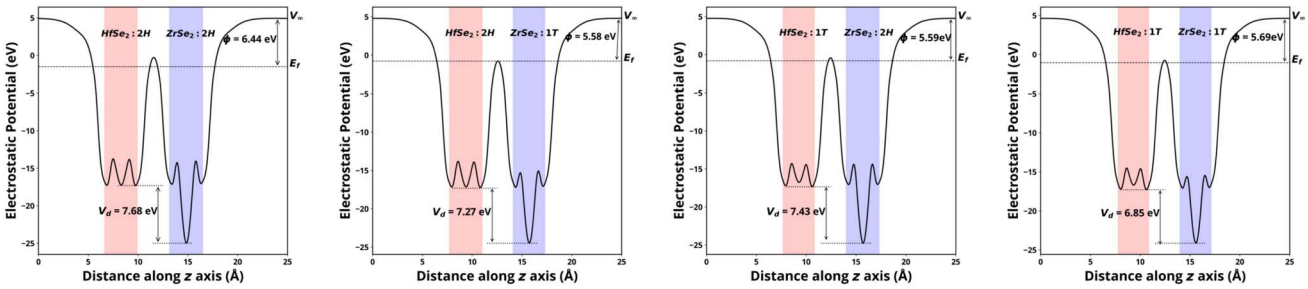
Fig. 7 Projected band structure of different heterostructure formed by considering the lowest energy. Red bands represent the $HfSe_2$ bands and blue bands represent $ZrSe_2$ bands.

lower CBM (M2) therefore M1 acts as an electron donor or donor material and M2 acts as an acceptor material. While on the other side, holes in the material with the lower VBM (M2) will move towards the material with the higher VBM (M1). This separation of electrons and holes reduces the likelihood of recombination at the interface and enhances the charge separation efficiency.

The power conversion efficiency in heterojunction solar cells, which is mainly determined by electron transfer between the monolayers, is a major factor in determining the proportion of solar energy that can be converted into electricity. This is the most crucial metric for the performance of a photovoltaic device. We have considered the approach proposed by Scharber[96] and Bernardi[97] to calculate the upper limit of the PCE of the 2D excitonic solar cells, in which the PCE is expressed as follows:

$$\eta = \frac{\beta_{\mathrm{FF}} V_{\mathrm{OC}} J_{\mathrm{SC}}}{P_{\mathrm{solar}}} = \frac{0.65\left(E_{\mathrm{g}}^{\mathrm{d}} - \Delta E_{\mathrm{c}} - 0.3\right) \int_{E_{\mathrm{g}}^{\mathrm{d}}}^{\infty} \frac{P(\hbar\omega)}{\hbar\omega}\, \mathrm{d}(\hbar\omega)}{\int_0^{\infty} P(\hbar\omega)\mathrm{d}(\hbar\omega)} \quad (8)$$

Here, $\beta_{\mathrm{FF}}$ is the fill factor taken to be 0.65, and $V_{\mathrm{OC}}$ is the maximum open-circuit voltage, which is determined as the




(a) $ZrSe_2$:2H/$HfSe_2$:2H (b) $ZrSe_2$:1T/$HfSe_2$:2H (c) $ZrSe_2$:2H/$HfSe_2$:1T (d) $ZrSe_2$:1T/$HfSe_2$:1T

Fig. 8 Electrostatic potential along the $z$ axis of the heterostructure, highlighting the work function $\phi$ and the electrostatic potential difference between the $HfSe_2$ and $ZrSe_2$ monolayers within the heterostructure.

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Table 4 Energy band gap of the heterostructure $E_g$; band gap of the donor $E_g^d$; conduction band offset $\Delta E_c$; efficiency of the device $\eta$; work function $\phi$; and electrostatic potential difference between the monolayers of the heterostructure $V_d$

| Heterostructure | $E_g$ (eV) | Band alignment | $E_g^d$ (eV) | $\Delta E_c$ (eV) | $\eta$ (%) | $\Phi$ (eV) | $V_d$ (eV) |
|---|---|---|---|---|---|---|---|
| $ZrSe_2$:2H/$HfSe_2$:2H | 1.23 | II | 1.28 | 0.05 | 21.95 | 6.44 | 7.68 |
| $ZrSe_2$:1T/$HfSe_2$:2H | 0.39 | II | 0.70 | 0.31 | 3.55 | 5.58 | 7.27 |
| $ZrSe_2$:2H/$HfSe_2$:1T | 0.35 | II | 0.93 | 0.58 | 1.68 | 5.59 | 7.43 |
| $ZrSe_2$:1T/$HfSe_2$:1T | 0.86 | I | 1.17 | 0.21 | — | 5.69 | 6.85 |

difference between the band gap of donor material ($E_g^d$) and the conduction band offset ($\Delta E_C$), which is shown in ESI Fig. 8,† subtracted by the energy loss due to the kinematics, which is taken to be 0.3 eV. These predictions were initially developed for studying bulk excitonic solar cells,[35,63–67,96] but later they were widely used to investigate promising 2D materials for excitonic photovoltaic devices.[97] The short-circuit current density ($J_{SC}$) is calculated assuming 100% external quantum efficiency using an integral that accounts for the energy flux of the AM1.5 solar spectrum. This integral considers the photon energy range from $E_g^d$ to infinity, and $P(\hbar\omega)$ represents the spectral power density of the solar spectrum as a function of photon energy ($\hbar\omega$). This is normalized over $P_{solar}$, which is calculated by integrating the entire AM1.5 solar spectrum ($P_{solar} \approx 1000$ W m$^{-2}$).

The $V_{OC}$ is determined as the energy difference between the VBM of the donor material and the CBM of the acceptor material. The larger energy difference between these is reflected in a higher $V_{OC}$, which leads to a higher PCE in solar cells. It is also worth noting that a higher $E_g^d$ reflects a higher $V_{OC}$; however, it reduces the $J_{SC}$ which lowers the PCE of the device. Therefore, $E_g^d$ should have an appropriate value ranging between 1 eV and around 1.7 eV. In this current work, one of the heterostructures shows type-I band alignment ($ZrSe_2$:1T/$HfSe_2$:1T), and the other three show type-II alignment, which is favorable for photovoltaic applications. The heterostructure formed by the combination of $ZrSe_2$:2H and $HfSe_2$:2H has a PCE of 21.95% which is shown in the contour plot in Fig. 10. The best PCE when using TMDs as monolayers has previously been found to be 21.43% – when $WTe_2$ and $PtSe_2$ are used as monolayers.[66] Our theoretically predicted efficiency exceeds this reported value, making the $ZrSe_2$:2H/$HfSe_2$:2H heterostructure a superior option for photovoltaics. Additionally, the obtained PCE is close to a previously reported photovoltaic heterostructure formed by MXenes, which had a PCE of 22.74%.[65] The $ZrSe_2$:2H layer, with a band gap of 1.28 eV in the $ZrSe_2$:2H/$HfSe_2$:2H heterostructure, efficiently absorbs a substantial portion of the incident solar radiation. A significantly lower band gap would not improve the $J_{SC}$ due to the minimal solar radiation available in that energy range. Moreover, a lower band gap would lead to a reduction in $V_{OC}$, which directly affects the overall efficiency ($\eta$). Additionally, the low conduction band offset (CBO) of 0.05 eV in this heterostructure further contributes to enhancing the $V_{OC}$. Although the other two combinations, $ZrSe_2$:1T/$HfSe_2$:2H and $ZrSe_2$:2H/$HfSe_2$:1T, exhibit type-II band alignment, they have a low band gap which favors enhancing $J_{SC}$ but greatly decreases $V_{OC}$. This is due to the effect of having a larger CBO, which consequently decreases the PCE.

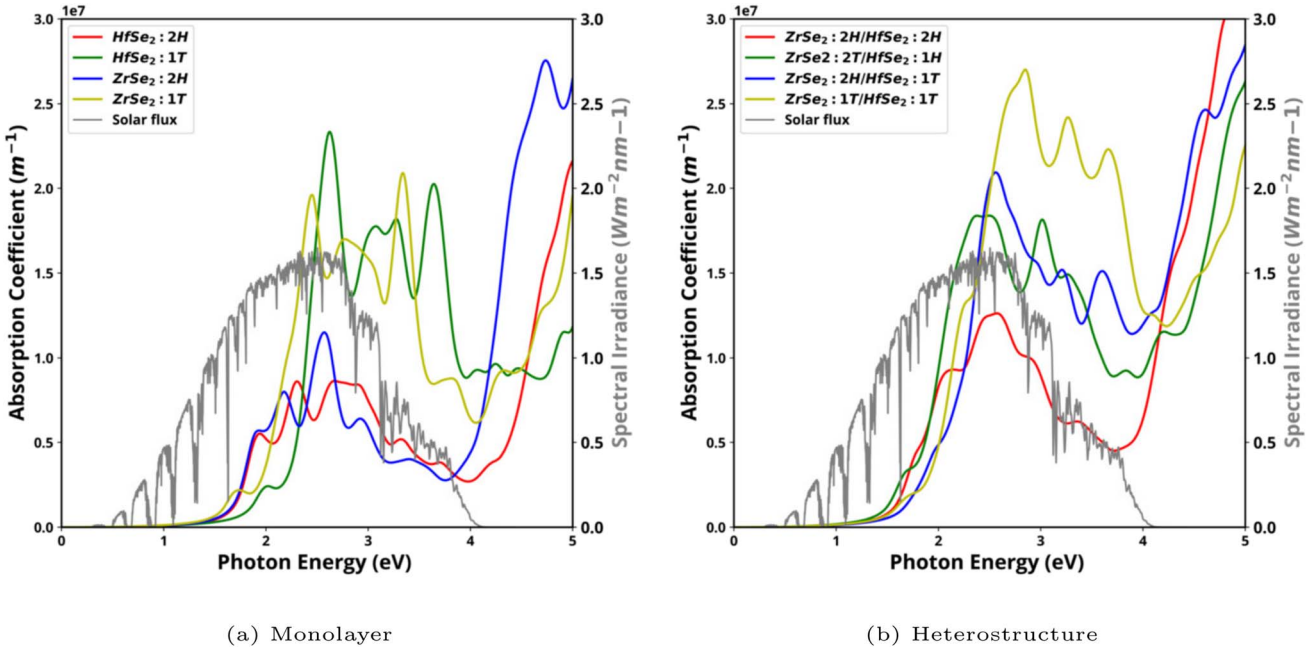


Fig. 9 Absorption spectra of monolayers and heterostructures shown in (a) and (b), respectively.

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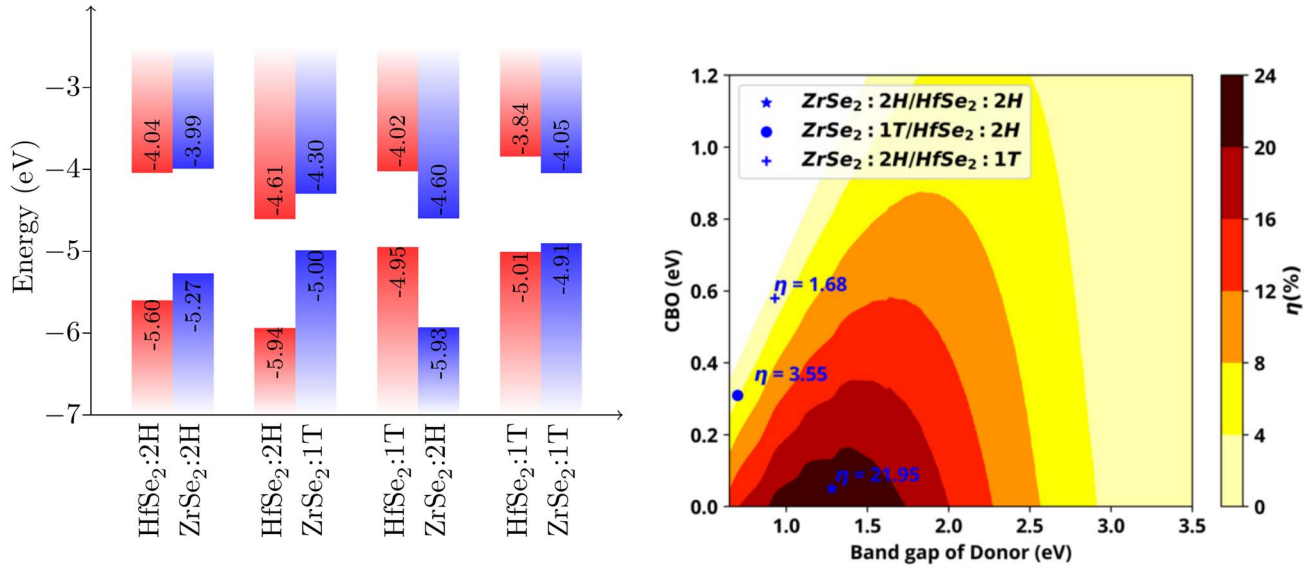


(a) Band edge alignment of monolayers in heterostructure

(b) Power Conversion Efficiency ($\eta$)

Fig. 10 (a) Band edge alignment of the monolayer within each heterostructure with respect to a vacuum reference and their band alignment. (b) Contour plot for power conversion efficiency $\eta$ as a function of band gap of the donor and CBO of the heterostructure, forming type-II band alignment.

## 4 Conclusions

The heterostructures formed by both 1T and 2H combinations of $ZrSe_2$ and $HfSe_2$ have been theoretically demonstrated using density functional theory calculations. By examining the formation energy, binding energy, mechanical stability, and dynamical stability, it has been determined that all monolayers and heterostructures are stable. According to the electronic band structure calculated using the HSE06 functional, the monolayer's band gap has an indirect band gap and falls within the range 0.97 eV to 1.53 eV, which is the optimum band gap for photovoltaics that can be used in the creation of heterostructures. Furthermore, the heterostructure generated harvests the majority of the visible part of the solar spectrum, as confirmed by the optical absorption spectra. Three out of the four heterostructures formed in the study show type-II band alignment, which help effectively separate the charge carriers, resulting in low carrier recombination rates. A significant electrostatic potential difference exists in the heterostructure between each monolayer, which facilitates the transfer of charge carriers between the layers by means of the internal electric field generated. The heterostructure, which is formed by $ZrSe_2$ and $HfSe_2$ has a moderate band gap that absorbs a larger area of the spectra and enhances $J_{SC}$. The heterostructure, which is constructed using the combination of both 2H phases of $ZrSe_2$ and $HfSe_2$, has a small conduction band offset of 0.05 eV, which helps to enhance the $V_{OC}$. Both the $J_{SC}$ and $V_{OC}$ are directly proportional to the efficiency, thus have a higher efficiency of 21.95%.

## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Conflicts of interest

The authors declare no conflict of interest.

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